अर्थ

ARTH

Gnani AI | Great Indian AI Internship Challenge 2026

The thinking behind Arth

A thumbprint proves who signed. Arth checks that the borrower understood.

Artificial intelligence is the tool. Real intelligence is the brain.

The problem

A thumbprint proves who signed a loan. It does not prove the borrower understood it. The RBI's Key Facts Statement circular (**RBI/2024-25/18**, 15 April 2024) asks lenders to give the KFS in a language the borrower understands, explain it, and take an acknowledgement that the borrower understood. For someone who cannot read, that acknowledgement can still be a mark on a page they cannot read. **Arth** (अर्थ, from the Sanskrit *artha*: meaning, and wealth) is built for that gap. It checks, out loud, whether the borrower understood what is on the paper, before they sign.

The idea

1 Everything is said out loud

Every screen speaks its own instruction in the borrower's language, with replay and slow buttons. Nobody has to read or type.

2 Teach-back, not tick-box

The borrower explains the loan in their own words. Open questions come in random order.

3 One engine, three doors

In person with the agent, online with the lender's KFS PDF or QR, and a call check on a synthetic sales call.

4 A shared receipt

A PDF of what was checked, with fingerprints of the paper and every clip. Not a signature, not a loan approval.

How it works

Paper terms PDF, QR or sample → **Agent reads aloud** Gnani Prisma hears → **Borrower explains** Timbre asks · Prisma hears → **Fixed rules** compare with the paper

✅ **Every answer matches** shared receipt, proof of understanding

🔁 **One answer differs** slow re-explain of that point, ask again

✋ **Three tries on one point** do not sign yet, bring someone you trust

The agent's words are checked too: "no processing fee" against a paper that says ₹1,500 is ❌ different. Results are matched, different, not mentioned or couldn't check, plus the question to ask. Fixed rules decide, not an AI model.

Why it is different

Who is checked

Many tools explain terms or screen calls.

Arth checks whether the **borrower understood**, in their own words, and whether the **agent said** what the paper says.

Who decides

Many tools let an AI model judge.

Fixed rules with unit tests decide. No AI judge, so every result traces back to evidence.

How it is used

Many tools assume reading and typing.

Zero reading: every screen speaks, buttons are large, no typing, and status is colour, icon and sound together.

Which models

Many tools mix several AI services.

Only Gnani: Prisma listens, Timbre speaks. Real calls, nothing faked.

What we researched

The rule behind the gap	The RBI Key Facts Statement circular (RBI/2024-25/18, 15 Apr 2024). It decided which terms we check: loan amount, money in hand, instalment and how often, processing fee, tenure and late fee.
The Gnani APIs, read and tested	Prisma takes at most 30 seconds per clip and reads the format from the file name, so the browser encodes its own 16 kHz WAV. Prisma v3.0 returns numbers as digits. Only one request runs at a time, so every call shares one queue with retry. Timbre needs numbers written as words.
A real Hindi and English test	A spoken sample in the Gnani playground showed that the Hindi "nahi" comes *after* the field name: आपको 30,000 का loan मिलेगा। EMI 2666 और processing fee नहीं लगेगी। That exact output became a unit-test fixture.
The other entries	We looked at what others in the challenge built, to be sure we were solving a different problem: understanding, not explaining or scam-spotting.
The law on voice	Voice is not a legal e-signature in India; recordings can be evidence, and hashes help. So the wording is always "proof of understanding, alongside the signature".
Who we designed for	Someone who may not read the paper at all, helped by an agent, a family member or an SHG *didi*. That is where the zero-reading design and the helper mode come from.

Consent, privacy and security, built in

✓ **Consent comes first.**	In agent mode the agent records a spoken consent line before anything about the loan.
✓ **Recording is always visible.**	Recording starts only on a tap, with a red pill and timer on screen. Nothing records in the background.
✓ **Every clip is labelled.**	Agent, borrower or helper, so it is clear who said what.
✓ **The borrower stays in control.**	Any answer can be said again. There is no typing anywhere.
✓ **One button wipes it.**	"सब मिटाएँ" clears the whole session. No login, no database: nothing is kept.
✓ **The key never leaves the server.**	The Gnani key lives only in server settings, never in the browser or in Git. Audio and document text are never logged.
✓ **Hardened against misuse.**	Rate limits and size caps on both voice routes, which accept only the models, voices and languages the app uses. Security headers on every page.
✓ **Not a signature.**	Voice is proof of understanding, alongside the signature.

Foul play we planned for

From the agent	Verbal mis-selling ("no fee"), coaching the answers, "the app says it's fine". Answered by the field-by-field agent check, random teach-back questions and no verdict words.
From either side	"Nobody explained", or a disputed recording. Answered by speaker labels, the agent's consent clip, and SHA-256 fingerprints of the paper and every clip on one receipt shared with both sides.
Out of scope, said openly	An agent and a borrower working together to fool the check. And Arth does not prove whose voice it heard.

Why it is built this way

Voice-first, zero reading	The borrower may not read. Every screen speaks, buttons are large, and status is colour plus icon plus sound: ✅ chime, ❌ buzz, ❔ grey.
Teach-back, not a quiz	Explaining in their own words shows more than tapping "yes". Open questions come in random order, including "what happens if you miss an instalment?"
No LLM decides anything	Money terms cannot be guessed. Fixed rules with **553 unit tests** make every comparison traceable to evidence.
Unknown stays unknown	A missing value is "not mentioned", never zero. A different unit or frequency (weekly vs monthly) is "couldn't check", never a conflict.
Money as pictures	Amounts are shown as notes and coins with digits (₹1,500 = 3 × ₹500), and spoken as words.
Careful words	Matched, different, not mentioned or couldn't check, plus the question to ask. Never "safe", "approved", "verified" or "fraud".

What we chose not to do

No AI judge · No OCR · No voice biometrics · No voice cloning · No face recognition · No APR unless validated · No real data · No legal-signature claim

Languages today

Hindi, English	Stable	Full flow, two voices each
Marathi	Under review	Selectable, wording being checked by a native speaker
Bengali, Gujarati, Kannada, Malayalam, Punjabi, Tamil, Telugu	Beta	Listed, not selectable yet

From plan to product: what changed, and why

LoanLens → Arth name	The plan was called LoanLens. The product became **Arth** (अर्थ): one word that means both "meaning" and "wealth", and is understood across many Indian languages (अर्थ · অর্থ · અર્થ).
Camera OCR dropped paper input	Reading a photo needs a third-party AI model. Arth reads **text-based KFS PDFs and a QR code** with plain code instead; "lenders print one QR on the KFS".
Teach-back made exact the gate	Checked **point by point**: money in hand, instalment and how often, fee, and what a missed instalment costs. A wrong number re-explains only that point; three tries on one point stops the flow.
More than Hindi languages	Hindi and English are stable, Marathi is under native review, and seven more Gnani languages are listed as Beta but cannot be selected until a native speaker checks the wording.
Instant, still real speed	Fixed prompts were generated once with **real Timbre calls** and stored, so screens speak at once. Anything else is spoken live; all listening is live Prisma.
Share to Arth phone	Installed on an Android phone, Arth appears in the share sheet, so a KFS PDF received on WhatsApp opens straight in the app.
Security pass before judging	Key and Git history audited (no leaks), voice routes locked to the app's own settings, test pages hidden, security headers added, dependencies patched.

Honest limits

A prototype	Accuracy has not been measured on real borrowers, so no accuracy figure is claimed. Hindi is the most tested flow. Arth does not prove whose voice it heard, and it checks only the listed terms. All data is synthetic, and the call check runs on a synthetic call only.